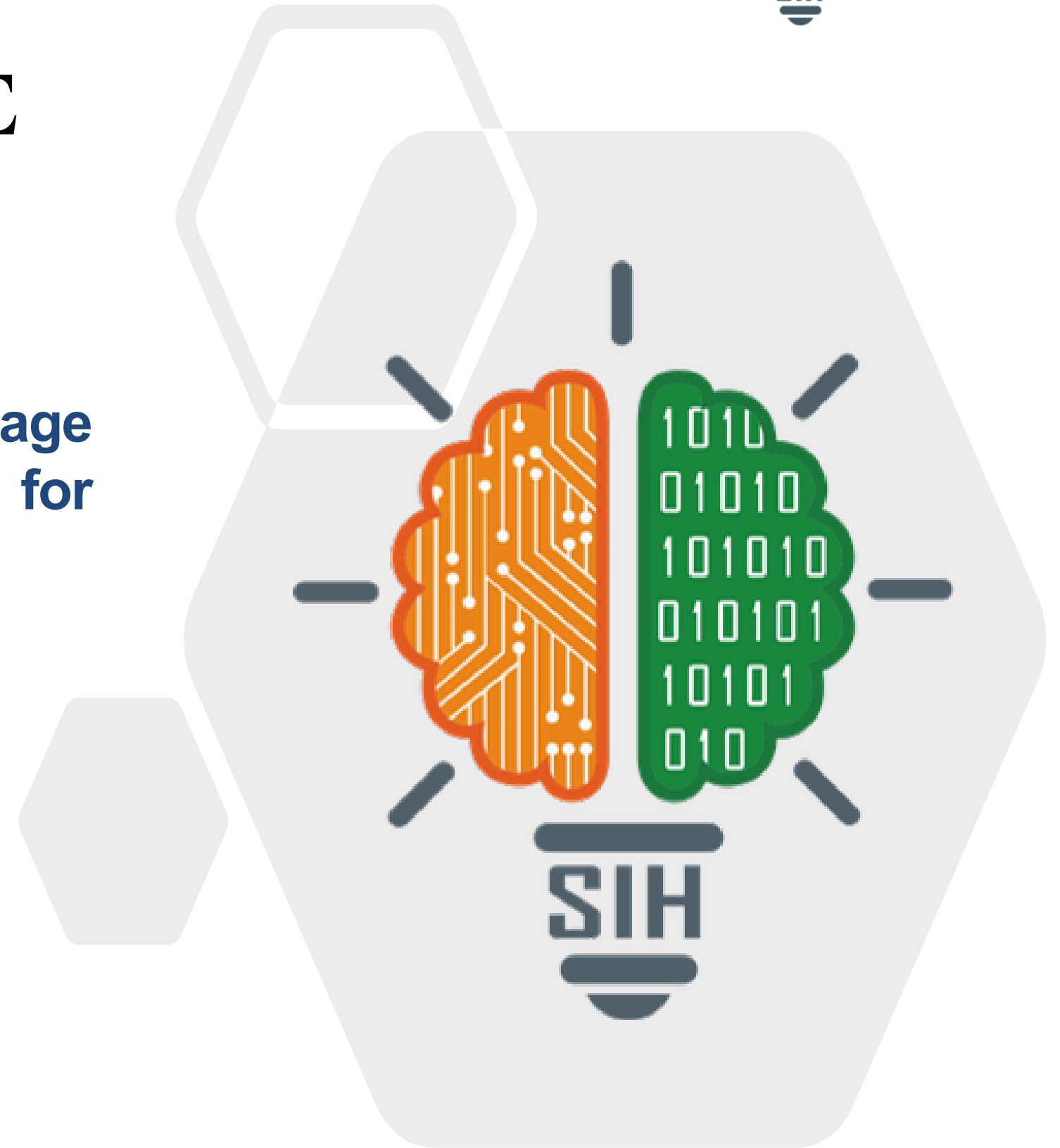


TITLE PAGE

- Problem Statement ID - 26090
- Problem Statement Title - AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans
- Theme - Heritage & Culture
- PS Category- Software
- Team ID -
- Team Name - Kuchu Puchu



THE PROBLEM

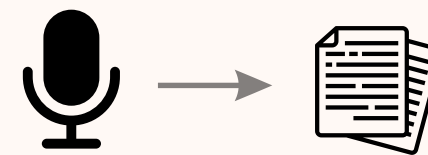
Skilled artisans create high-quality products but face major barriers in selling online. Language, product photography, catalog creation, pricing, and SEO make digital commerce difficult, limiting their access to wider markets and customers.

THE SOLUTION

A voice-first, multilingual digital commerce platform designed for low literacy artisans. The artisan simply speaks about the product and uploads a photo; the platform transforms it into a professional, market-ready product listing.

OUR PLATFORM

Smart Content Engine



- Speech-to-text
- Keyword extractbn
- Product title
- Description

Smart Image Engine



- Background removal
- Blur reduction
- Lighting correction
- Image upscaling

Smart Pricing Engine



- Minimum price protection
- Market demand analysis
- Recommended price
- Dynamic price range

Regional Language → Hindi / English



GUIDED COMMUNICATION FOR BULK ORDERS



B2B

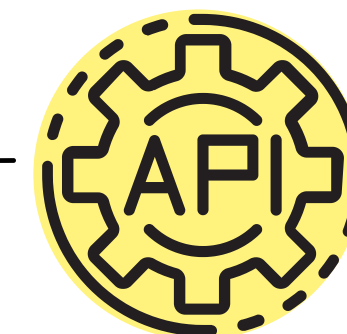
Buy in bulk for business needs



Government E- Marketplace



Marketplace-Ready Listing



Github

KARIGAR Digital Commerce Architecture



WHY THIS WORKS

- **Easy for Artisans** : Voice and image-based product onboarding reduces digital complexity.
- **Multilingual Access** : Regional language support removes language barriers.
- **AI-Powered Cataloging** : Automatically creates professional titles, descriptions and product details.
- **Better Product Presentation** : Image enhancement improves listing quality.
- **Smarter Pricing** : Market data helps suggest a practical price range.
- **Marketplace Ready** : Final output is structured for direct digital commerce use.

IMPLEMENTATION FLOW

React + JavaScript

Frontend UI for artisan, buyer and analytics interfaces.

Python

Core language for AI, image processing and pricing logic.

OpenCV/ Pillow

Performs image pre-processing, correction and enhancement.

Qwen2.5-instruc

Generates structured product titles, descriptions, features and keywords.

Bhashini

Regional-language speech recognition and multilingual translation.

Express.js

Handles core application logic, APIs, orders and buyer-artisan interactions.

FastAPI

Exposes Python-based AI/ML services through lightweight APIs.

Rembg

Removes product-image backgrounds automatically.

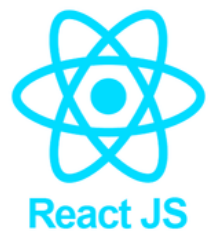
Qwen2.5-VL

Extracts visual product attributes and understands uploaded product images.

PostgreSQL

Stores users, products, prices, orders, enquiries and analytics data.

TECHNOLOGY STACK



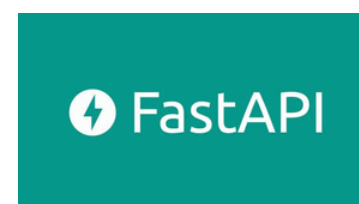
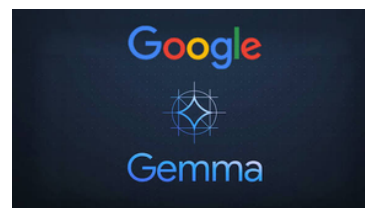
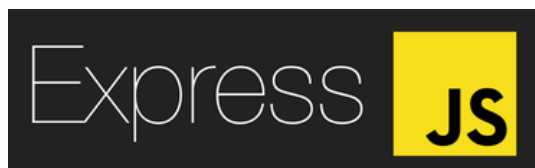
React JS



JavaScript



python™



TECHNICAL FEASIBILITY

- Proven AI Stack: Pre-trained Indic STT/translation, Computer Vision & ML models no foundational model training required.
- Modular Architecture: Flutter/React + FastAPI + PostgreSQL; independent Voice, Vision & Pricing services.
- Cross-Platform & Cloud-Ready: Single ecosystem across Web, Android & iOS with scalable cloud deployment.
- Human-in-the-Loop: Artisan validation before listing publication ensures output accuracy.

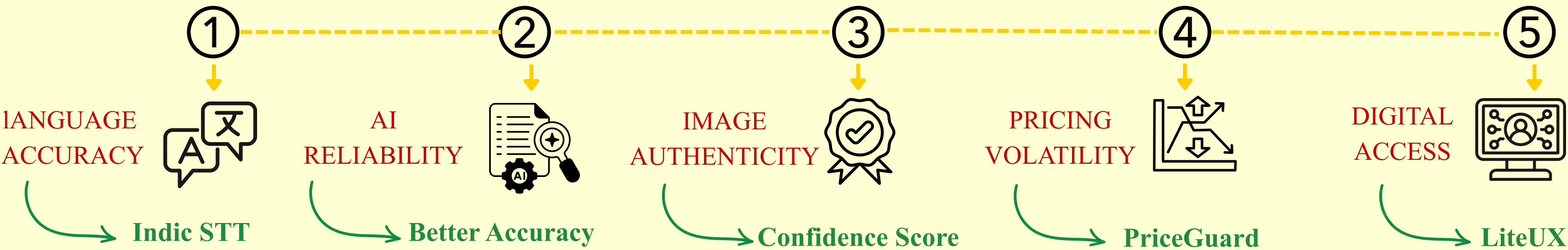
BUSINESS POTENTIAL

- Dual Market Access: B2C retail + B2B/bulk procurement within a unified platform.
- Asset-Light Model: No inventory ownership; technology-led marketplace enablement.
- Revenue Potential: Transaction commission + B2B facilitation / value-added services.
- Scalable User Base:
 - Individual Artisans
 - SHGs/Clusters
 - Institutional onboarding
 - Pan-India expansion.

OPERATIONAL VIABILITY

- Voice-First UX: Minimal typing & digital-literacy dependency
- Vernacular-First: Regional-language interaction with Hindi/English buyer-side translation
- Guided Workflows: Structured listing, pricing & B2B interactions
- Low Adoption Barrier: Mobile-first access with minimal onboarding/training

CHALLENGE TO MITIGATION



IMPACT

- Assisted Onboarding Completion
- B2B Enquiry-to-Proposal Conversion
- Successful Vernacular Interaction Rate
- Improvement in price realisation measured against artisan baseline/local selling price.

Category

Key Features and Benefits



ARTISAN

- Zero-Skill Digital Selling
- Minimum Price Protection
- Unified Buyer Access
- Business Visibility



BUYER & MARKET

- Data-Backed Pricing
- Multilingual Commerce
- Structured Bulk Ordering
- Formal Procurement Access



ECOSYSTEM & ECONOMY

- Digitised Informal Commerce
- Cluster-Level Scalability
- Handicraft Export Enablement

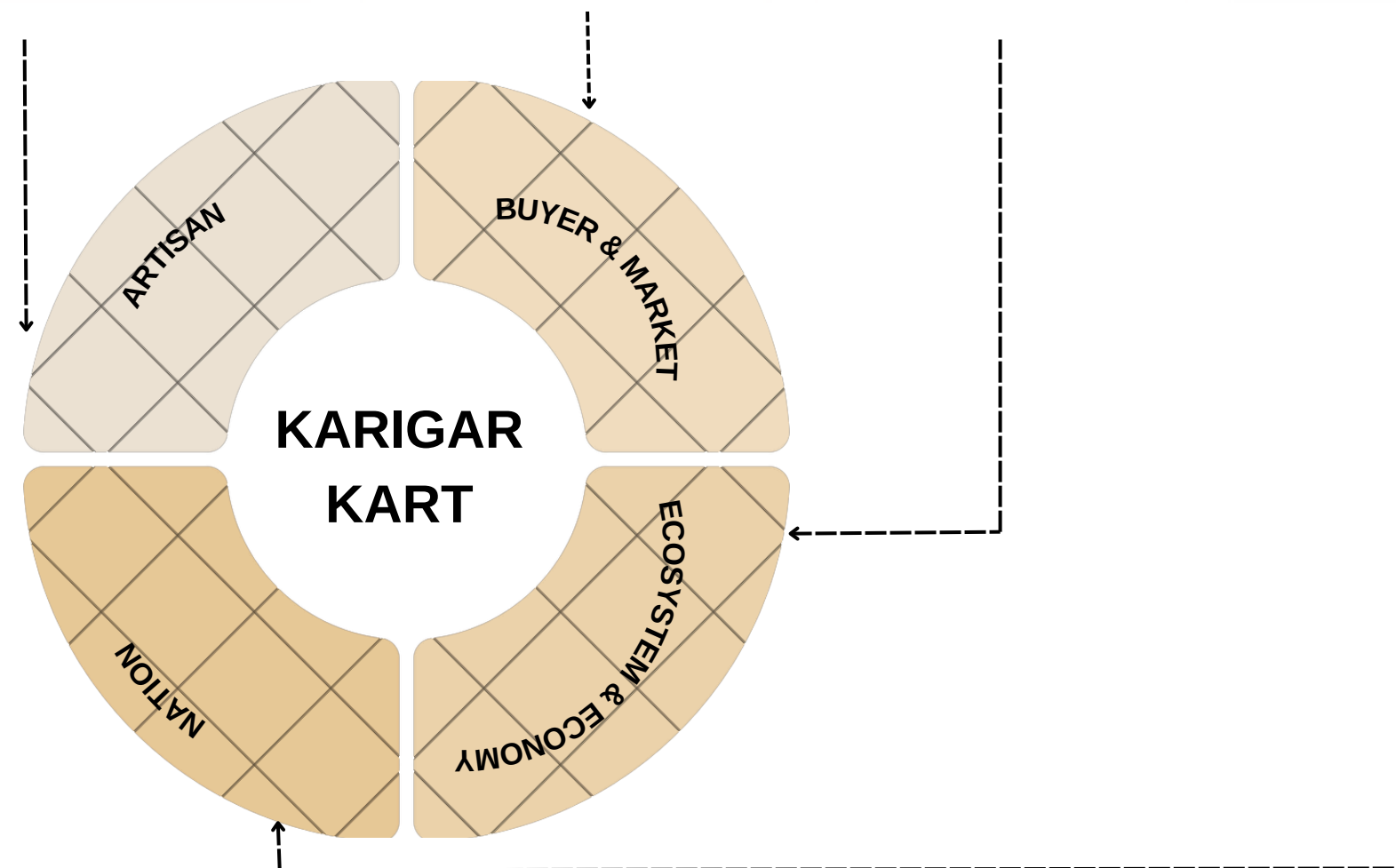


NATION

- Digital India Alignment
- Vocal for Local
- Skill India Support
- GeM Procurement Enablement
- SDG 8 & SDG 5 Alignment

QUANTIFIABLE KPIs

- Reduction in listing time voice + automated content/image processing versus manual listing.
- Listing completion rate target adoption among low-digital-literacy artisan users.
- Improvement in product discoverability measured through product views/enquiries after catalogue optimisation.
- Increase in buyer reach expansion from local/offline buyers to B2C + B2B digital channels.



Technology	Description	System Usage	Reference
Speech-to-Text: Bhashini	Multilingual speech recognition models convert regional-language voice input into text.	Converts artisan voice notes into Hindi/English text for catalog generation.	Radford et al., BHASHINI – Digital India Language Technology Initiative https://bhashini.gov.in/gyankosh?tab=publications
Image Enhancement: Real-ESRGAN + OpenCV + rembg	Real-ESRGAN enhances low-quality images, while rembg automatically removes backgrounds and opencv pillow do denoising add brightness and contrast	Converts poor mobile-camera product photos into clean, marketplace-ready catalog images.	Real-ESRGAN Paper https://arxiv.org/abs/2107.10833 rembg Open-Source Documentation https://arxiv.org/html/2308.09764v2 https://github.com/danielgatis/rembg/blob/main/README.md
Vision-Language Model: Qwen2.5-VL-3B	Multimodal model capable of image understanding and structured product-information extraction.	Identifies product category, material, color, design, style, and other attributes from images.	Bai et al., <i>Qwen2.5-VL Technical Report</i> , 2025 https://arxiv.org/abs/2502.13923
Catalog Generation: Qwen2.5-Instruct	Instruction-tuned multilingual LLM for structured text generation.	Generates professional product titles, descriptions, tags, keywords, and SEO-friendly content.	Yang et al., <i>Qwen2.5 Technical Report</i> , 2024 https://huggingface.co/Qwen/Qwen2.5-7B-Instruct
Market Search: SerpAPI	Retrieves structured Google Search/Shopping results through an API.	Finds similar products and collects comparable market information and prices.	SerpAPI – Google Search API Documentation https://serpapi.com/google-shopping-api
Price Recommendation: LightGBM / XGBoost	Gradient-boosting regression algorithms capable of learning relationships between product and market features.	Uses product attributes and comparable market prices to generate a recommended selling-price range.	Ke et al., <i>LightGBM: A Highly Efficient Gradient Boosting Decision Tree</i> , NeurIPS 2017, XGBoost: A Scalable Tree Boosting System
Database: PostgreSQL / Supabase	Relational database infrastructure for structured application data.	Stores artisan profiles, product catalogs, generated descriptions, market data, and pricing results.	PostgreSQL / Supabase Official Documentation https://github.com/postgres/postgres
Backend API: FastAPI	High-performance Python framework for developing REST APIs.	Connects the frontend with AI models, image-processing pipeline, pricing engine, and database.	FastAPI Official Documentation https://github.com/fastapi